

Assignment 15

1. Create a Login program that throws an exception for invalid password and uses finally block.

Code:

```
import java.util.Scanner;
```

```
public class LoginException {
```

```
    static void login(String password) throws Exception {  
        if (!password.equals("12345")) {  
            throw new Exception("Invalid password!");  
        }  
    }
```

```
        System.out.println("Login successful.");  
    }
```

```
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        try {  
            System.out.print("Enter password: ");  
            String password = sc.nextLine();
```

```
            login(password);
```

```
        } catch (Exception e) {  
            System.out.println("Error: " + e.getMessage());
```

```
        } finally {  
            System.out.println("Login process completed.");  
        }
```

```
        sc.close();
```

```
}  
}
```

Output:

```
Enter password: 123  
Error: Invalid password!  
Login process completed.
```

For correct password:

```
Enter password: 12345  
Login successful.  
Login process completed.
```

2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

Code:

```
import java.util.Scanner;
```

```
public class ATM_PIN {
```

```
    static void verifyPIN(int pin) throws Exception {  
        if (pin != 1234) {  
            throw new Exception("Invalid PIN!");  
        }  
    }
```

```
        System.out.println("PIN verified successfully.");  
    }
```

```
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        try {  
            System.out.print("Enter ATM PIN: ");
```

```
        int pin = sc.nextInt();

        verifyPIN(pin);

    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    } finally {
        System.out.println("PIN verification process completed.");
    }

    sc.close();
}
```

Output:

```
Enter ATM PIN: 5678
Error: Invalid PIN!
PIN verification process completed.
```

Correct PIN:

```
Enter ATM PIN: 1234
PIN verified successfully.
PIN verification process completed.
```